CSE 644 Homework1

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1. Install Docker

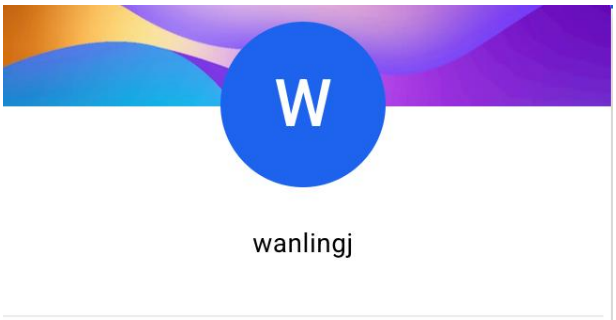
```
[vagrant@vagrant:~$ docker --version
Docker version 29.6.2, build dfc4efb
[vagrant@vagrant:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:c3cbe1cc1aa588a64951ac6286e0df7b27fe2e6324b1001c619bb358770c0178
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.
```

2. Create Docker Hub Account

My Docker Hub username: wanlingj

Docker Hub profile link: https://hub.docker.com/u/wanlingj



3. Generate Credentials for Docker Image Upload

Copy access token

Use this token as a password when you sign in from the Docker CLI client. Learn more

Make sure you copy your personal access token now. Your personal access token is only displayed once. It isn't stored and can't be retrieved later.

Access token description
CSE644 Docker Assignment Upload

Expires on
Never

Access permissions
Read, Write, Delete

To use the access token from your Docker CLI client:

1. Run

```
$  docker login -u wanlingj
```

Copy

2. At the password prompt, enter the personal access token.

```
d[redacted]4
```

Copy

Back to access tokens

```
vagrant@vagrant:~$ docker login -u wanlingj

i Info → A Personal Access Token (PAT) can be used instead.
         To create a PAT, visit https://app.docker.com/settings


Password:

WARNING! Your credentials are stored unencrypted in '/home/vagrant/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
```

4. Pull, Run, and Exec into a Docker Image

```
vagrant@vagrant:~$ docker pull ubuntu:24.04
24.04: Pulling from library/ubuntu
ca2678b20700: Pull complete
55378bbcdb75: Download complete
Digest: sha256:4fbb8e6a8395de5a7550b33509421a2bafbc0aab6c06ba2cef9ebffbc7092d90
Status: Downloaded newer image for ubuntu:24.04
docker.io/library/ubuntu:24.04
```

```
vagrant@vagrant:~$ docker images

IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   c3cbe1cc1aa5       25.9kB         9.49kB    U
ubuntu:24.04         4fbb8e6a8395        119MB         31.7MB
```

```
vagrant@vagrant:~$ docker run -d --name ubuntu-demo ubuntu:24.04 sleep infinity
1b6e1f8387f55b4db00b4df67bcefc87c5f2c0c6c42ba2529dd4292b0ac041cd
vagrant@vagrant:~$ docker ps
CONTAINER ID   IMAGE          COMMAND            CREATED          STATUS          PORTS     NAMES
1b6e1f8387f5   ubuntu:24.04   "sleep infinity"   32 seconds ago   Up 31 seconds             ubuntu-demo
```

```
vagrant@vagrant:~$ docker exec -it ubuntu-demo /bin/bash
root@1b6e1f8387f5:/#
root@1b6e1f8387f5:/# hostname
1b6e1f8387f5
```

5. Customize an Existing Web Server Image

```
vagrant@vagrant:~/cse644-web-custom$ cat Dockerfile
# 基于官方Nginx基础镜像
FROM nginx:latest

# 将本地自定义网页复制到Nginx站点根目录，覆盖默认页面
COPY index.html /usr/share/nginx/html/index.html

# 对外暴露80网页端口
EXPOSE 80

# Nginx前台启动命令（容器标准启动指令）
CMD ["nginx", "-g", "daemon off;"]
vagrant@vagrant:~/cse644-web-custom$ cat index.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>CSE644 Custom Nginx Web Page</title>
</head>
<body>
    <h1>Cloud Computing CSE644 Docker Assignment</h1>
    <h2>Customized Nginx Docker Image Demo</h2>
    <p>Student: wanlingj</p>
    <p>This webpage is embedded inside a custom Docker Nginx image.</p>
</body>
</html>
```

```
vagrant@vagrant:~/cse644-web-custom$ docker build -t wanlingj/cse644-custom-nginx:v1 .
[+] Building 16.0s (6/7)                                                                          docker:default
 => [internal] load build definition from Dockerfile                                                       0.0s
 => => transferring dockerfile: 348B                                                                       0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                                            6.4s
 => [auth] library/nginx:pull token for registry-1.docker.io                                         => [auth]
 library/nginx:pull token for registry-1.docker.io                                         => [auth] library/n
ginx:pull token for registry-1.docker.io                                         => [auth] library/nginx:pull
token for registry-1.docker.io                                                        0.0s
 => [internal] load .dockerignore                                                                          0.0s
 => => transferring context: 2B                                                                            0.0s
 => [internal] load build context                                                                          0.0s
 => => transferring context: 388B                                                                          0.0s
 => [1/2] FROM docker.io/library/nginx:latest@sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942   9.5s
[+] Building 17.2s (8/8) FINISHED                                                                docker:default. =
> [internal] load build definition from Dockerfile                                                       0.0s
 => => transferring dockerfile: 348B                                                                      0.0s. =
> [internal] load metadata for docker.io/library/nginx:latest                                            6.4s
 => [auth] library/nginx:pull token for registry-1.docker.io                                              0.0s. =
> [internal] load .dockerignore                                                                          0.0s
 => => transferring context: 2B                                                                           0.0s. =
> [internal] load build context                                                                          0.0s
 => => transferring context: 388B                                                                         0.0s. =
> [1/2] FROM docker.io/library/nginx:latest@sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942   9.7s
 => => resolve docker.io/library/nginx:latest@sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942   0.0s. =
> => sha256:d26f27cc8c41e321394cb3c9a80915d90d5f1f1d3cbbbcda3be00f13c53b041e 1.40kB / 1.40kB                 0.1s
 => => sha256:062e450697faa5f02a3a74eba9864ee4d79bc9cfbd65769fc6cdff2c05c6a053 29.78MB / 29.78MB            6.7s. =
> => sha256:b6698f04e005497a7f495c0719358d43890cb3997ad7b4ab0b06748247c574a3 403B / 403B                     0.5s
 => => sha256:2bedaf25031a24fb70b9dc2d56cb17139186d1ae5fd2054ecbd0dfe1a69585ba 1.21kB / 1.21kB              0.5s. =
> => sha256:cacfcdd01f309c65d69372716e799ea741065ac1b1e60880b3a6981ae105cb55 955B / 955B                     0.4s
 => => sha256:3c7ab7949321f47c96fc0918f9f72e8f51bd452cdef1e0dad9599880317380b9 626B / 626B                  0.1s. =
> => sha256:82454cdbf456a77f9ff1bb88b121c2a739e38c30ea689c135c7cca6249eabe4e 33.33MB / 33.33MB               7.0s
 => => extracting sha256:062e450697faa5f02a3a74eba9864ee4d79bc9cfbd65769fc6cdff2c05c6a053                   1.5s. =
> => extracting sha256:82454cdbf456a77f9ff1bb88b121c2a739e38c30ea689c135c7cca6249eabe4e                     1.1s
 => => extracting sha256:3c7ab7949321f47c96fc0918f9f72e8f51bd452cdef1e0dad9599880317380b9                   0.0s
 => => extracting sha256:cacfcdd01f309c65d69372716e799ea741065ac1b1e60880b3a6981ae105cb55                   0.0s
 => => extracting sha256:b6698f04e005497a7f495c0719358d43890cb3997ad7b4ab0b06748247c574a3                   0.0s
 => => extracting sha256:2bedaf25031a24fb70b9dc2d56cb17139186d1ae5fd2054ecbd0dfe1a69585ba                   0.0s
 => => extracting sha256:d26f27cc8c41e321394cb3c9a80915d90d5f1f1d3cbbbcda3be00f13c53b041e                   0.0s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html                                                 0.7s
 => exporting to image                                                                                     0.2s
 => => exporting layers                                                                                    0.1s
 => => exporting manifest sha256:f8c3606598c2e350adc3779f86088b69a06a9565821bd3fbbb1c112bba430ae8          0.0s
 => => exporting config sha256:b718dfd463dbc8b90b0d7314f4ec5bf8df7471b89e1a2ea46a37d8f5c4daab2b             0.0s
 => => exporting attestation manifest sha256:89b29a735e63cfd87efc58b1627235c8e6f545b669be8ff42f2d381ac356c32c   0.0s
 => => exporting manifest list sha256:1fc0930dc1d3ae71d05e75e1e077c08e0f214e3661c65a7a6603d9ab1669fbb5     0.0s
 => => naming to docker.io/wanlingj/cse644-custom-nginx:v1                                                 0.0s
 => => unpacking to docker.io/wanlingj/cse644-custom-nginx:v1                                              0.0s
vagrant@vagrant:~/cse644-web-custom$
```

```
vagrant@vagrant:~/cse644-web-custom$ docker images

IMAGE                            ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest               c3cbe1cc1aa5       25.9kB         9.49kB     U
ubuntu:24.04                     4fbb8e6a8395        119MB         31.7MB     U
wanlingj/cse644-custom-nginx:v1  1fc0930dc1d3        238MB         63.1MB
```

```
vagrant@vagrant:~/cse644-web-custom$ docker run -d --name custom-web -p 8080:80 wanlingj/cse644-custom-nginx:v1
8d41b8dd132869bf6058afbf7ebe6d02e3a7f84a2ee3be81c45865ca4999f660
vagrant@vagrant:~/cse644-web-custom$ docker ps
CONTAINER ID   IMAGE                             COMMAND                  CREATED          STATUS          PORTS                                     NAMES
8d41b8dd1328   wanlingj/cse644-custom-nginx:v1   "/docker-entrypoint.…"   22 seconds ago   Up 21 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   custom-web
1b6e1f8387f5   ubuntu:24.04                      "sleep infinity"         21 minutes ago   Up 21 minutes                                             ubuntu-demo
```

```
vagrant@vagrant:~/cse644-web-custom$ curl localhost:8080
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>CSE644 Custom Nginx Web Page</title>
</head>
<body>
    <h1>Cloud Computing CSE644 Docker Assignment</h1>
    <h2>Customized Nginx Docker Image Demo</h2>
    <p>Student: wanlingj</p>
    <p>This webpage is embedded inside a custom Docker Nginx image.</p>
</body>
</html>
```

6. Create a Simple Python or Node.js Web Server Image

```
vagrant@vagrant:~/cse644-python-web$ cat app.py
from http.server import HTTPServer, BaseHTTPRequestHandler

class Handler(BaseHTTPRequestHandler):
    def do_GET(self):
        self.send_response(200)
        self.send_header("Content-type", "text/html")
        self.end_headers()
        html = """
        <h1>CSE644 Python Web Server</h1>
        <p>Running on port 8888 inside Docker Container</p>
        <p>Student: wanlingj</p>
        """
        self.wfile.write(html.encode())

# 固定监听0.0.0.0:8888
if __name__ == "__main__":
    server = HTTPServer(("0.0.0.0", 8888), Handler)
    print("Web server listening on port 8888")
    server.serve_forever()
```

```
vagrant@vagrant:~/cse644-python-web$ cat Dockerfile
# 官方轻量Python基础镜像
FROM python:3.11-slim

# 设置容器工作目录
WORKDIR /app

# 拷贝本地Python脚本进容器
COPY app.py /app/

# 暴露要求的8888端口
EXPOSE 8888

# 容器启动命令，运行web服务
CMD ["python3", "app.py"]
```

```
docker build -t wanlingj/cse644-python-web:v1 .
[+] Building 5.4s (9/9) FINISHED                                                                                              docker:default
 => [internal] load build definition from Dockerfile                                                                                    0.0s
 => => transferring dockerfile: 297B                                                                                                    0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                                                     1.4s
 => [auth] library/python:pull token for registry-1.docker.io                                                                           0.0s
 => [internal] load .dockerignore                                                                                                       0.0s
 => => transferring context: 2B                                                                                                         0.0s
 => [1/3] FROM docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c7627accd3d51053a93               3.2s
 => => resolve docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c7627accd3d51053a93               0.1s
 => => sha256:c89b9f64c028c19ba92e195d7589d914b9cd1fc69f3c3dfeb931e93876ac2064 248B / 248B                                              0.1s
 => => sha256:6b265b8eae4a26a263eb55546aa430c1eef37032a7cd86393d98c6b779f28319 14.42MB / 14.42MB                                        2.2s
 => => sha256:9775d166087aba0afe5ba9a88859c04c08090e82ae218742d09fec9119e0335d 1.29MB / 1.29MB                                          1.3s
 => => extracting sha256:9775d166087aba0afe5ba9a88859c04c08090e82ae218742d09fec9119e0335d                                               0.2s
 => => extracting sha256:6b265b8eae4a26a263eb55546aa430c1eef37032a7cd86393d98c6b779f28319                                               0.8s
 => => extracting sha256:c89b9f64c028c19ba92e195d7589d914b9cd1fc69f3c3dfeb931e93876ac2064                                               0.0s
 => [internal] load build context                                                                                                       0.1s
 => => transferring context: 656B                                                                                                       0.0s
 => [2/3] WORKDIR /app                                                                                                                  0.2s
 => [3/3] COPY app.py /app/                                                                                                             0.0s
 => exporting to image                                                                                                                  0.3s
 => => exporting layers                                                                                                                 0.1s
 => => exporting manifest sha256:0f3d529792769e54c809749eb8f2d3af7aa43e6e3ae67187677fa676c5c43667                                       0.0s
 => => exporting config sha256:2657c7285bfad48eb69ae0e2b4610b30c24780c6ed419aa626b7da829b5574e6                                         0.0s
 => => exporting attestation manifest sha256:f329c2d5550a7b2181c128191e927e4d996aa366684c9bd0b8ec241de0172bac                           0.0s
 => => exporting manifest list sha256:f9f8c89067cacffcb0ded4dba12c807704649c127bf97d89cc71f1723d728abf                                  0.0s
 => => naming to docker.io/wanlingj/cse644-python-web:v1                                                                                0.0s
 => => unpacking to docker.io/wanlingj/cse644-python-web:v1                                                                             0.1s
```

```
vagrant@vagrant:~/cse644-python-web$ docker run -d --name python-web-demo -p 0.0.0.0:8888:8888 wanlingj/cse644-python-web:v1
4d436f521312bb557f0058974e5ca976d7a3e741d58f8930683097414997a74d
vagrant@vagrant:~/cse644-python-web$ docker ps
CONTAINER ID   IMAGE                             COMMAND                  CREATED          STATUS          PORTS                                     NAMES
4d436f521312   wanlingj/cse644-python-web:v1     "python3 app.py"         16 seconds ago   Up 16 seconds   0.0.0.0:8888->8888/tcp                    python-web-demo
8d41b8dd1328   wanlingj/cse644-custom-nginx:v1   "/docker-entrypoint.…"   21 minutes ago   Up 21 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   custom-web
1b6e1f8387f5   ubuntu:24.04                      "sleep infinity"         42 minutes ago   Up 42 minutes                                             ubuntu-demo
```

```
vagrant@vagrant:~/cse644-python-web$ ss -tulpn
Netid         State          Recv-Q         Send-Q                    Local Address:Port                   Peer Address:Port        Process
udp           UNCONN         0              0                            127.0.0.54:53                          0.0.0.0:*
udp           UNCONN         0              0                         127.0.0.53%lo:53                          0.0.0.0:*
udp           UNCONN         0              0                        10.0.2.15%eth0:68                          0.0.0.0:*
tcp           LISTEN         0              4096                            0.0.0.0:8080                        0.0.0.0:*
tcp           LISTEN         0              4096                         127.0.0.54:53                          0.0.0.0:*
tcp           LISTEN         0              4096                      127.0.0.53%lo:53                          0.0.0.0:*
tcp           LISTEN         0              4096                            0.0.0.0:22                          0.0.0.0:*
tcp           LISTEN         0              4096                            0.0.0.0:8888                        0.0.0.0:*
tcp           LISTEN         0              4096                               [::]:8080                           [::]:*
tcp           LISTEN         0              4096                               [::]:22                             [::]:*
```

```
vagrant@vagrant:~/cse644-python-web$ curl localhost:8888

    <h1>CSE644 Python Web Server</h1>
    <p>Running on port 8888 inside Docker Container</p>
    <p>Student: wanlingj</p>
```

7. Deploy HAProxy as a Proxy to Nginx

```
vagrant@vagrant:~/cse644-haproxy-nginx$ docker ps
CONTAINER ID   IMAGE                             COMMAND                  CREATED              STATUS              PORTS                                     NAMES
c4c6620555f1   haproxy:latest                    "docker-entrypoint.s…"   49 seconds ago       Up 49 seconds       0.0.0.0:80->80/tcp                        haproxy-proxy
2e20608331e8   wanlingj/cse644-custom-nginx:v1   "/docker-entrypoint.…"   About a minute ago   Up About a minute   80/tcp                                    web2
58946a18df28   wanlingj/cse644-custom-nginx:v1   "/docker-entrypoint.…"   About a minute ago   Up About a minute   80/tcp                                    web1
4d436f521312   wanlingj/cse644-python-web:v1     "python3 app.py"         15 minutes ago       Up 15 minutes       0.0.0.0:8888->8888/tcp                    python-web-demo
8d41b8dd1328   wanlingj/cse644-custom-nginx:v1   "/docker-entrypoint.…"   36 minutes ago       Up 36 minutes       0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   custom-web
1b6e1f8387f5   ubuntu:24.04                      "sleep infinity"         57 minutes ago       Up 57 minutes                                             ubuntu-demo
```

We can see that web1, web2 and haproxy-proxy are running.

```
vagrant@vagrant:~/cse644-haproxy-nginx$ cat haproxy.cfg
global
    maxconn 2000
    daemon
defaults
    mode http
    timeout connect 5s
    timeout client  30s
    timeout server  30s
# 前端监听80端口，接收外部请求
frontend http_front
    bind *:80
    default_backend nginx_servers
# 后端负载均衡组，指向两台Nginx容器
backend nginx_servers
    server web1 web1:80 check
    server web2 web2:80 check
```

```
docker exec web1 cat /usr/share/nginx/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>CSE644 Custom Nginx Web Page</title>
</head>
<body>
    <h1>Cloud Computing CSE644 Docker Assignment</h1>
    <h2>Customized Nginx Docker Image Demo</h2>
    <p>Student: wanlingj</p>
    <p>This webpage is embedded inside a custom Docker Nginx image.</p>
</body>
</html>
```

```
vagrant@vagrant:~/cse644-haproxy-nginx$ curl localhost:80
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>CSE644 Custom Nginx Web Page</title>
</head>
<body>
    <h1>Cloud Computing CSE644 Docker Assignment</h1>
    <h2>Customized Nginx Docker Image Demo</h2>
    <p>Student: wanlingj</p>
    <p>This webpage is embedded inside a custom Docker Nginx image.</p>
</body>
</html>
```

```
vagrant@vagrant:~/cse644-haproxy-nginx$ docker logs haproxy-proxy
[NOTICE]   (1) : Initializing new worker (9)
[NOTICE]   (1) : Loading success.
```

8. Demonstrate Persistent Volume

```
vagrant@vagrant:~$ docker volume create cse644-data-volume
cse644-data-volume
vagrant@vagrant:~$ docker volume ls
DRIVER    VOLUME NAME
local     cse644-data-volume
```

```
vagrant@vagrant:~$ docker run -it --name volume-demo -v cse644-data-volume:/mnt/data ubuntu:24.04 /bin/bash
root@f3f1ef1184b6:/# echo "CSE644 Persistent Volume Test Data - Student wanlingj" > /mnt/data/persist-demo.txt
root@f3f1ef1184b6:/# cat /mnt/data/persist-demo.txt
CSE644 Persistent Volume Test Data - Student wanlingj
root@f3f1ef1184b6:/# exit
exit
```

```
vagrant@vagrant:~$ docker rm volume-demo
volume-demo
vagrant@vagrant:~$ docker ps -a
CONTAINER ID   IMAGE                              COMMAND                  CREATED              STATUS                   PORTS                                     NAMES
c4c6620555f1   haproxy:latest                     "docker-entrypoint.s…"   16 minutes ago       Up 16 minutes            0.0.0.0:80->80/tcp                        haproxy-proxy
2e20608331e8   wanlingj/cse644-custom-nginx:v1    "/docker-entrypoint.…"   16 minutes ago       Up 16 minutes            80/tcp                                    web2
58946a18df28   wanlingj/cse644-custom-nginx:v1    "/docker-entrypoint.…"   16 minutes ago       Up 16 minutes            80/tcp                                    web1
4d436f521312   wanlingj/cse644-python-web:v1      "python3 app.py"         31 minutes ago       Up 31 minutes            0.0.0.0:8888->8888/tcp                    python-web-demo
8d41b8dd1328   wanlingj/cse644-custom-nginx:v1    "/docker-entrypoint.…"   52 minutes ago       Up 52 minutes            0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   custom-web
1b6e1f8387f5   ubuntu:24.04                       "sleep infinity"         About an hour ago    Up About an hour                                                   ubuntu-demo
8e22b319ee2e   hello-world                        "/hello"                 2 hours ago          Exited (0) 2 hours ago                                             nervous_murdock
```

We can see that docker ps -a shows no volume-demo.

```
vagrant@vagrant:~$ docker run -it --name volume-demo-new -v cse644-data-volume:/mnt/data ubuntu:24.04 /bin/bash
root@4e24e74e283c:/# cat /mnt/data/persist-demo.txt
CSE644 Persistent Volume Test Data - Student wanlingj
```

We can see when we run a new container, the data still exits.

```
vagrant@vagrant:~$ docker volume inspect cse644-data-volume
[
    {
        "CreatedAt": "2026-07-19T05:24:49Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/cse644-data-volume/_data",
        "Name": "cse644-data-volume",
        "Options": null,
        "Scope": "local"
    }
]
```

9. Demonstrate Docker Networking

We create two bridges, web-bridge and db-bridge:

```
vagrant@vagrant:~$ docker network create web-bridge
5b546bb25ccb2fe67b3eac0c890d203587e4609357f00a5d765a01cfeb62d53c
vagrant@vagrant:~$ docker network create db-bridge
de541de0a82591195e3186a4d307017e6ea414517a2643af94094bf60fbde9ed
vagrant@vagrant:~$ docker network ls
NETWORK ID     NAME         DRIVER    SCOPE
bcf7f6e24b44   bridge       bridge    local
de541de0a825   db-bridge    bridge    local
642f054e751e   host         host      local
e8eb3a720f4a   none         null      local
bf2405266430   proxy-net    bridge    local
5b546bb25ccb   web-bridge   bridge    local
```

We create 4 containers:

Container a and b belong to web-bridge, container c belong to db-bridge, container d is separeted from web-bridge and db-bridge:

```
vagrant@vagrant:~$ docker run -d --name web-a --network web-bridge ubuntu:24.04 sleep infinity
docker run -d --name web-b --network web-bridge ubuntu:24.04 sleep infinity
71e854fe943fad7f99baf1ef04d2581caff17a3929e796f7fb9380878849c855
edcf4a1c800f2f066c235f5eca6f5bbb6e06593cac478c8267cec2e283906e07
vagrant@vagrant:~$ docker run -d --name db-c --network db-bridge ubuntu:24.04 sleep infinity
06c7108aa20a886815508d8d3b80a401303c34b5cf7fe08de8be77c7aa214df6
vagrant@vagrant:~$ docker run -d --name default-d ubuntu:24.04 sleep infinity
19d5d65efbb41e502da75fccaba80cf824600dd98be873ec26e5d397e8b05019
```

web-a can ping web-b:

```
vagrant@vagrant:~$ docker exec -it web-a ping -c 3 web-b
PING web-b (172.19.0.3) 56(84) bytes of data.
64 bytes from web-b.web-bridge (172.19.0.3): icmp_seq=1 ttl=64 time=0.087 ms
64 bytes from web-b.web-bridge (172.19.0.3): icmp_seq=2 ttl=64 time=0.059 ms
64 bytes from web-b.web-bridge (172.19.0.3): icmp_seq=3 ttl=64 time=0.081 ms

--- web-b ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2084ms
rtt min/avg/max/mdev = 0.059/0.075/0.087/0.012 ms
```

web-a cannot ping web-c or default-d:

```
vagrant@vagrant:~$ docker exec -it web-a ping -c 3 db-c
ping: db-c: Temporary failure in name resolution
vagrant@vagrant:~$ docker exec -it web-a ping -c 3 default-d
ping: default-d: Temporary failure in name resolution
```

```
vagrant@vagrant:~$ docker network inspect web-bridge
[
    {
        "Name": "web-bridge",
        "Id": "5b546bb25ccb2fe67b3eac0c890d203587e4609357f00a5d765a01cfeb62d53c",
        "Created": "2026-07-19T05:39:59.312805739Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.19.0.0/16",
                    "Gateway": "172.19.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {},
        "Labels": {},
        "Containers": {
            "71e854fe943fad7f99baf1ef04d2581caff17a3929e796f7fb9380878849c855": {
                "Name": "web-a",
                "EndpointID": "f32e6b49ca68f3a7e9cf7b1706d5d761742fa0e50f435011a206dbab798e4c36",
                "MacAddress": "c6:a2:4f:be:d0:33",
                "IPv4Address": "172.19.0.2/16",
                "IPv6Address": ""
            },
            "edcf4a1c800f2f066c235f5eca6f5bbb6e06593cac478c8267cec2e283906e07": {
                "Name": "web-b",
                "EndpointID": "b7e728660e46e9e1ae21b5162de237fb35f6350a662c277e1ac0869dd57287a0",
                "MacAddress": "fa:67:2c:ab:2f:8b",
                "IPv4Address": "172.19.0.3/16",
                "IPv6Address": ""
            }
        },
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.19.0.0/16": {
                        "IPsInUse": 5,
                        "DynamicIPsAvailable": 65531
                    }
                }
            }
        }
    }
]
```

Host network:

```
vagrant@vagrant:~$ docker run -d --name host-demo --network host ubuntu:24.04 bash -c "apt update && apt install -y iproute2 curl iputils-ping; sleep infinity"
3bde03dba209e6f475ce01cff02214f2c2e03fabc5c57931e82d6aa30e9e6732
```

```
vagrant@vagrant:~$ docker exec -it host-demo ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f8:c2:eb brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic eth0
       valid_lft 77526sec preferred_lft 77526sec
    inet6 fd17:625c:f037:2:a00:27ff:fef8:c2eb/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86042sec preferred_lft 14042sec
    inet6 fe80::a00:27ff:fef8:c2eb/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 92:80:4d:f9:7a:a1 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
    inet6 fe80::9080:4dff:fef9:7aa1/64 scope link
       valid_lft forever preferred_lft forever
6: vethe337eac@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether 3e:0b:66:8b:02:b2 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::3c0b:66ff:fe8b:2b2/64 scope link
       valid_lft forever preferred_lft forever
7: veth40bc01a@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether e6:a2:c1:7d:e1:a5 brd ff:ff:ff:ff:ff:ff link-netnsid 1
    inet6 fe80::e4a2:c1ff:fe7d:e1a5/64 scope link
       valid_lft forever preferred_lft forever
8: vethce6085f@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether d6:79:82:c6:20:dc brd ff:ff:ff:ff:ff:ff link-netnsid 2
    inet6 fe80::d479:82ff:fec6:20dc/64 scope link
       valid_lft forever preferred_lft forever
9: br-bf2405266430: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether ea:fc:ee:fb:95:e9 brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-bf2405266430
       valid_lft forever preferred_lft forever
    inet6 fe80::e8fc:eeff:fefb:95e9/64 scope link
       valid_lft forever preferred_lft forever
10: veth36fc548@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-bf2405266430 state UP group default
    link/ether 1e:d0:2f:27:b7:01 brd ff:ff:ff:ff:ff:ff link-netnsid 3
    inet6 fe80::1cd0:2fff:fe27:b701/64 scope link
       valid_lft forever preferred_lft forever
11: veth45c32cd@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-bf2405266430 state UP group default
    link/ether 46:90:32:fb:13:f7 brd ff:ff:ff:ff:ff:ff link-netnsid 4
    inet6 fe80::4490:32ff:fefb:13f7/64 scope link
       valid_lft forever preferred_lft forever
12: veth12030e4@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-bf2405266430 state UP group default
    link/ether 1a:45:01:59:c6:0e brd ff:ff:ff:ff:ff:ff link-netnsid 5
    inet6 fe80::1845:1ff:fe59:c60e/64 scope link
       valid_lft forever preferred_lft forever
17: br-5b546bb25ccb: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 76:71:20:aa:b1:c2 brd ff:ff:ff:ff:ff:ff
```


```
17: br-5b546bb25ccb: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 76:71:20:aa:b1:c2 brd ff:ff:ff:ff:ff:ff
```


```
    inet 172.19.0.1/16 brd 172.19.255.255 scope global br-5b546bb25ccb
       valid_lft forever preferred_lft forever
    inet6 fe80::7471:20ff:feaa:b1c2/64 scope link
       valid_lft forever preferred_lft forever
18: br-de541de0a825: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 5e:b7:5f:71:f4:8f brd ff:ff:ff:ff:ff:ff
    inet 172.20.0.1/16 brd 172.20.255.255 scope global br-de541de0a825
       valid_lft forever preferred_lft forever
    inet6 fe80::5cb7:5fff:fe71:f48f/64 scope link
       valid_lft forever preferred_lft forever
19: veth5680b47@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-5b546bb25ccb state UP group default
    link/ether d6:f9:eb:ca:b5:61 brd ff:ff:ff:ff:ff:ff link-netnsid 6
    inet6 fe80::d4f9:ebff:feca:b561/64 scope link
       valid_lft forever preferred_lft forever
20: veth22d9d49@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-5b546bb25ccb state UP group default
    link/ether 16:9e:5c:30:bb:09 brd ff:ff:ff:ff:ff:ff link-netnsid 7
    inet6 fe80::149e:5cff:fe30:bb09/64 scope link
       valid_lft forever preferred_lft forever
21: vethf2b850b@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-de541de0a825 state UP group default
    link/ether ca:63:5a:14:3d:bb brd ff:ff:ff:ff:ff:ff link-netnsid 8
    inet6 fe80::c863:5aff:fe14:3dbb/64 scope link
       valid_lft forever preferred_lft forever
22: veth23f66fc@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether 16:4e:d7:97:65:65 brd ff:ff:ff:ff:ff:ff link-netnsid 9
    inet6 fe80::144e:d7ff:fe97:6565/64 scope link
       valid_lft forever preferred_lft forever
```

In host mode, container can access host 8888:

```
vagrant@vagrant:~$ docker exec -it host-demo curl localhost:8888

        <h1>CSE644 Python Web Server</h1>
        <p>Running on port 8888 inside Docker Container</p>
        <p>Student: wanlingj</p>
```

Isolated network:

```
vagrant@vagrant:~$ docker run -it --name none-demo --network none ubuntu:24.04 /bin/bash
root@67b5228d8adf:/# ip a
bash: ip: command not found
root@67b5228d8adf:/# cat /proc/net/dev
Inter-|   Receive                                                |  Transmit
 face |bytes    packets errs drop fifo frame compressed multicast|bytes    packets errs drop fifo colls carrier compressed
    lo:       0       0    0    0    0     0          0         0        0       0    0    0    0     0       0          0
root@67b5228d8adf:/#
```

Only lo exists, there is no eth0 or anything else.

10. Upload Created or Modified Images to Docker Hub

```
vagrant@vagrant:~$ docker login
Authenticating with existing credentials... [Username: wanlingj]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
```

```
vagrant@vagrant:~$ docker images

IMAGE                               ID             DISK USAGE   CONTENT SIZE   EXTRA
haproxy:latest                      728fec0da177        176MB         49.2MB    U
hello-world:latest                  c3cbe1cc1aa5       25.9kB         9.49kB    U
ubuntu:24.04                        4fbb8e6a8395        119MB         31.7MB    U
wanlingj/cse644-custom-nginx:v1     1fc0930dc1d3        238MB         63.1MB    U
wanlingj/cse644-nginx:v1            1fc0930dc1d3        238MB         63.1MB    U
wanlingj/cse644-python-web:v1       f9f8c89067ca        187MB         45.5MB    U
```

```
vagrant@vagrant:~$ docker push wanlingj/cse644-nginx:v1
The push refers to repository [docker.io/wanlingj/cse644-nginx]
062e450697fa: Mounted from wanlingj/cse644-custom-nginx
82454cdbf456: Mounted from wanlingj/cse644-custom-nginx
cacfcdd01f30: Mounted from wanlingj/cse644-custom-nginx
2bedaf25031a: Mounted from wanlingj/cse644-custom-nginx
b342b840d57c: Mounted from wanlingj/cse644-custom-nginx
4e4eea52e99f: Mounted from wanlingj/cse644-custom-nginx
3c7ab7949321: Mounted from wanlingj/cse644-custom-nginx
b6698f04e005: Mounted from wanlingj/cse644-custom-nginx
d26f27cc8c41: Mounted from wanlingj/cse644-custom-nginx
v1: digest: sha256:1fc0930dc1d3ae71d05e75e1e077c08e0f214e3661c65a7a6603d9ab1669fbb5 size: 856
vagrant@vagrant:~$ docker push wanlingj/cse644-python-web:v1
The push refers to repository [docker.io/wanlingj/cse644-python-web]
0b7a01417a60: Pushed
062e450697fa: Mounted from wanlingj/cse644-nginx
9775d166087a: Pushed
6b265b8eae4a: Pushed
c89b9f64c028: Pushed
0d96cb6eba3d: Pushed
81c08a8769f6: Pushed
v1: digest: sha256:f9f8c89067cacffcb0ded4dba12c807704649c127bf97d89cc71f1723d728abf size: 856
```

```
vagrant@vagrant:~$ docker images

IMAGE                               ID             DISK USAGE   CONTENT SIZE   EXTRA
haproxy:latest                      728fec0da177        176MB         49.2MB    U
hello-world:latest                  c3cbe1cc1aa5       25.9kB         9.49kB    U
ubuntu:24.04                        4fbb8e6a8395        119MB         31.7MB    U
wanlingj/cse644-custom-nginx:v1     1fc0930dc1d3        238MB         63.1MB    U
wanlingj/cse644-haproxy:v1          1213ed0bba25        174MB         47.1MB
wanlingj/cse644-nginx:v1            1fc0930dc1d3        238MB         63.1MB    U
wanlingj/cse644-python-web:v1       f9f8c89067ca        187MB         45.5MB    U
```

```
vagrant@vagrant:~$ docker push wanlingj/cse644-haproxy:v1
The push refers to repository [docker.io/wanlingj/cse644-haproxy]
6378ed450c2e: Mounted from library/haproxy
062e450697fa: Mounted from wanlingj/cse644-python-web
de02cab19962: Mounted from library/haproxy
2e7f03d18c25: Mounted from library/haproxy
6db2f0514811: Mounted from library/haproxy
4f4fb700ef54: Mounted from library/haproxy
2f30a18f201c: Pushed
6cca4297ecd2: Pushed
v1: digest: sha256:1213ed0bba253bfea245d1546b7a6201f4cf58613ef2ff7fe598c41f67133316 size: 856
```

wanlingj/cse644-nginx：

https://hub.docker.com/repository/docker/wanlingj/cse644-nginx/general

wanlingj/cse644-python-web：

https://hub.docker.com/repository/docker/wanlingj/cse644-python-web/general

wanlingj/cse644-haproxy：

https://hub.docker.com/repository/docker/wanlingj/cse644-haproxy/general

11. Create or Use a GitHub Account

My GitHub account name is wanlingj-cse

12. Upload Scripts and Source Files to GitHub

GitHub repository link:

https://github.com/wanlingj-cse/cse644-docker-assignment/tree/master

```
vagrant@vagrant:~$ git push -f origin master
Username for 'https://github.com': wanlingj-cse
Password for 'https://wanlingj-cse@github.com':
Enumerating objects: 37, done.
Counting objects: 100% (37/37), done.
Delta compression using up to 2 threads
Compressing objects: 100% (28/28), done.
Writing objects: 100% (37/37), 6.93 KiB | 2.31 MiB/s, done.
Total 37 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:      https://github.com/wanlingj-cse/cse644-docker-assignment/pull/new/master
remote:
To https://github.com/wanlingj-cse/cse644-docker-assignment.git
 * [new branch]      master -> master
```